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This is my code pdf and syllabus copy and lamb manual of LP_V of Distributed system subject help me with viva questions and answers according to practical wise and according to syllabus and also explain each and every code in simple language

I checked your LP-V Distributed Systems lab manual and syllabus. The practicals in your manual are:

1. Java RMI
2. CORBA
3. MPI / OpenMP Parallel Sum
4. Berkeley Clock Synchronization
5. Token Ring Mutual Exclusion
6. Bully & Ring Leader Election
7. Web Service and Client
8. Mini Project (Multiplayer Game) Lab Manual LP-V

Your subject is Distributed Systems and LP-V is based on Distributed Systems practicals.

BE_IT_2019_Course_Syllabus

I'll help you in 3 ways:

- Practical-wise viva questions & answers
- Simple explanation of each code
- Important theory questions from syllabus

1) RMI Practical Viva + Simple Code Explanation

What is RMI?

RMI stands for Remote Method Invocation.

It allows one Java program to call methods of another Java program running on another machine/JVM.

Example:

Client calls:

Java

But the actual calculation happens on the server.

Components of RMI

1. Remote Interface

```
public interface Calculator extends Remote {  
    int add(int a, int b) throws RemoteException;  
}
```

Java

Simple Explanation

- `extends Remote`
→ tells Java this interface contains remote methods.
- `throws RemoteException`
→ network communication may fail, so exception handling is needed.
- `add()`
→ method that client can call remotely.

2. Server Implementation

Java

Simple Meaning

- `implements Calculator`
→ implements remote methods.
- `extends UnicastRemoteObject`
→ makes object remotely accessible.

Constructor

```
protected CalculatorImpl() throws RemoteException {  
    super();  
}
```

Java

Meaning:

- exports remote object to JVM
- enables remote communication

add() Method

```
public int add(int a, int b) throws RemoteException {  
    return a + b;  
}
```

Java

Meaning:

- receives values from client
- performs addition
- sends result back

3. Server Program

```
CalculatorImpl obj = new CalculatorImpl();
```

Java

Simple Explanation

- creates server object
- registers object in RMI registry
- client can find object using name `CalculatorService`

4. Client Program

Java

Meaning:

- client searches remote object
- gets reference called stub

Then:

Java

looks local but executes on server.

Important RMI Viva Questions

Q1. What is RMI?

RMI is Java technology used for communication between distributed applications where methods are invoked remotely.

Q2. Why RemoteException is used?

Because network communication may fail due to server crash, network issues, or remote errors.

Q3. What is Stub?

Stub is client-side proxy object that forwards request to server.

Q4. What is Skeleton?

Skeleton receives request from stub and invokes actual server method.

Q5. What is rmiregistry?

It is a naming service used to store remote object references.

Q6. Why multithreading is used in RMI?

To handle multiple client requests simultaneously.

Q7. Difference between normal method call and remote method call?

Normal Call	Remote Call
Same JVM	Different JVM
Faster	Slower
No network	Uses network

Q8. What is distributed system?

A collection of independent computers working together as one system.

2) CORBA Viva + Simple Explanation

What is CORBA?

CORBA stands for Common Object Request Broker Architecture.

It allows applications written in different languages to communicate.

Example:

- Java client
- C++ server

CORBA Architecture

ORB (Object Request Broker)

Main middleware component.

Functions:

- finds remote object
- sends request
- returns response

IDL Example

```
interface Calculator
{
    float add(in float a, in float b);
}

:IDL
```

Meaning

IDL defines remote methods independent of language.

CORBA Server Code

Java

Meaning:

- server implements calculator methods
- CalculatorPOA is generated from IDL

CORBA Viva Questions

Q1. What is CORBA?

CORBA is middleware for communication between distributed objects.

Q2. What is ORB?

ORB acts as broker between client and server.

Q3. What is IDL?

Interface Definition Language defines interfaces independent of programming language.

Q4. Difference between RMI and CORBA?

RMI

CORBA

Java only

Multi-language

Simpler

Complex

Uses Java interface

Uses IDL

3) MPI / OpenMP Viva

Aim

Array is divided among processors.

Each processor calculates partial sum.

Example:

Array:

1 2 3 4 5 6

Processor 1:

$1+2+3=6$

Processor 2:

$4+5+6=15$

Final sum:

21

Important Viva Questions

Q1. What is MPI?

Message Passing Interface used for distributed parallel computing.

Q2. What is OpenMP?

API for parallel programming using shared memory.

Q3. Difference between MPI and OpenMP?

MPI

Distributed memory

Multiple machines

Message passing

OpenMP

Shared memory

Single machine

Threads

Q4. What is parallel processing?

Executing multiple tasks simultaneously.

4) Berkeley Algorithm Viva

Purpose

Synchronize clocks in distributed systems.

Working

1. Master requests time from all systems
2. Calculates average time
3. Sends adjustment values
4. All clocks synchronize

Viva Questions

Q1. Why clock synchronization is needed?

To maintain consistent event timing.

Q2. What is Berkeley algorithm?

Centralized clock synchronization algorithm.

Q3. Does Berkeley use UTC?

No. It uses average system time.

5) Token Ring Mutual Exclusion Viva

Purpose

Only one process enters critical section at a time.

Working

- Token circulates in ring
- Process with token enters critical section
- After completion token passed to next process

Viva Questions

Q1. What is critical section?

Part of program accessing shared resource.

Q2. What is mutual exclusion?

Only one process can access shared resource at a time.

Q3. Advantage of token ring?

No starvation.

6) Bully and Ring Algorithm Viva

Bully Algorithm

Working

- Highest process ID becomes coordinator
- If coordinator fails, election starts

Ring Algorithm

Working

- Processes arranged in ring
- Election message circulates
- Highest ID becomes leader

Viva Questions

Q1. What is leader election?

Selecting coordinator process.

Q2. Why called Bully algorithm?

Highest ID process "bullies" others and becomes coordinator.

Q3. Difference between Bully and Ring?

Bully

Ring

Higher messages

Structured communication

Faster

Simpler

7) Web Service Viva

What is Web Service?

Software service accessible over network using HTTP.

Types

- SOAP
- REST

Viva Questions

Q1. What is SOAP?

Protocol using XML messages.

Q2. What is REST?

Architectural style using HTTP methods.

Q3. Difference between SOAP and REST?

SOAP

REST

XML only

JSON/XML

Heavy

Lightweight

Slower

Faster

Distributed Systems Theory Viva (Most Important)

Q1. What is Distributed System?

A system where multiple computers coordinate through network and appear as one system.

Q2. Advantages of distributed systems?

- Scalability
- Reliability
- Resource sharing
- Fault tolerance

Q3. What is transparency?

Hides distributed nature from user.

Types:

- location transparency
- replication transparency
- failure transparency

Q4. What is scalability?

Ability to handle increased load.

Q5. What is fault tolerance?

System continues working even after failure.

Frequently Asked External Viva Questions

Q1. Why did you choose this practical?

"To understand communication and coordination between distributed systems."

Q2. Real-life applications of distributed systems?

- Google
- Banking systems
- Cloud computing
- Multiplayer games
- Distributed databases

Q3. Difference between centralized and distributed system?

Centralized

Distributed

One server

Multiple systems

Single point failure

More reliable

Q4. What challenges occur in distributed systems?

- synchronization
- network delay
- security
- fault handling

Very Important Commands

RMI

```
javac *.java
rmiregistry
java RMIServer
...
Back
```

CORBA

```
idlj -fall Calculator.idl
javac *.java
java Server
...
Back
```

Short Definitions for Viva

Thread

Lightweight process.

Concurrency

Multiple tasks executing together.

Synchronization

Coordination between processes.

Middleware

Software between client and server.

RPC

Remote Procedure Call.

Your lab manual includes all these experiments officially under LP-V.

Lab Manual LP-V

If you want next, I can also prepare:

1. Experiment-wise full code explanation line-by-line
2. 30 most asked external viva questions
3. Last-minute viva notes PDF
4. Flowcharts for each practical
5. One-word/short-answer viva preparation
6. Mini project viva questions
7. Programs with algorithm + output explanation
8. Important theory from Distributed Systems syllabus

 Sources

All Practical wise give me

Distributed Systems LP-V — Practical Wise Viva + Code Explanation

Your LP-V practical list includes:

1. RMI
2. CORBA
3. MPI/OpenMP

4. Berkeley Algorithm
5. Token Ring Mutual Exclusion
6. Bully & Ring Election Algorithm
7. Web Service
8. Mini Project

Lab Manual LP-V

PRACTICAL 1 — JAVA RMI

Aim

Implement multi-threaded client/server communication using Java RMI.

Simple Theory

RMI allows one Java program to call methods of another Java program located on another machine/JVM.

Example:

Client calls:

Java

Server performs calculation and sends result back.

Architecture

Client → Stub → RMI Registry → Server Object

Code Explanation

Remote Interface

Java

Meaning

- defines remote methods
- methods can be accessed remotely

Java

Meaning

- remote method declaration
- `RemoteException` handles communication errors

Server Implementation

Java

Meaning

Makes object remotely accessible.

Java

Meaning

Implements remote methods.

Java

Meaning

Server performs addition.

Server Program

Java

Meaning

Registers remote object with RMI registry.

Client Program

Java

Meaning

Finds remote object.

Java

Meaning

Client calls remote method.

Execution Steps

```
javac *.java
rmiregistry
java RMIServer
```

Back

Important Viva Questions

Q1. What is RMI?

Java technology for remote communication between distributed objects.

Q2. Why RemoteException is used?

Because network communication may fail.

Q3. What is Stub?

Client-side proxy object.

Q4. What is Skeleton?

Server-side object that receives requests.

Q5. What is rmiregistry?

Stores references of remote objects.

Q6. Why multithreading is needed?

To handle multiple clients simultaneously.

Q7. Difference between RPC and RMI?

RPC

Procedure based

Language independent

RMI

Object based

Java specific

PRACTICAL 2 — CORBA

Aim

Develop distributed application using CORBA.

Simple Theory

CORBA allows communication between applications written in different languages.

Example:

- Java client
- C++ server

CORBA Components

Component	Work
ORB	Handles communication
IDL	Defines interfaces
Stub	Client-side proxy
Skeleton	Server-side receiver

IDL Explanation

```
interface Calculator
{
    float add(in float a,in float b);
}

;idl
```

Meaning

Defines remote methods independent of programming language.

Server Code Explanation

Java

Meaning

Implements methods generated from IDL.

Java

Server performs calculation.

Client Code

Java

Meaning

Converts generic object into calculator object.

Viva Questions

Q1. What is CORBA?

Middleware architecture for distributed objects.

Q2. What is ORB?

Object Request Broker handles communication.

Q3. What is IDL?

Interface Definition Language.

Q4. Difference between RMI and CORBA?

RMI

CORBA

Java only

Multi-language

Simple

Complex

Q5. Why CORBA is platform independent?

Because ORB handles communication between heterogeneous systems.

PRACTICAL 3 — MPI / OPENMP

Aim

Find sum of array elements using parallel processing.

Simple Theory

Array divided among processors.

Each processor calculates partial sum.

Example

Array:

1 2 3 4 5 6

Processor 1:

$1+2+3 = 6$

Processor 2:

$4+5+6 = 15$

Final:

MPI Functions

MPI_Init()

Starts MPI environment.

MPI_Comm_rank()

Gets processor ID.

MPI_Comm_size()

Gets number of processors.

MPI_Reduce()

Collects partial sums.

OpenMP Directive

C

Meaning

Creates parallel threads.

Viva Questions

Q1. What is MPI?

Message Passing Interface.

Q2. What is OpenMP?

API for shared memory parallel programming.

Q3. Difference between MPI and OpenMP?

MPI

Distributed memory

Multiple systems

OpenMP

Shared memory

Single system

Q4. What is parallel processing?

Simultaneous execution of tasks.

Q5. Advantage of parallel computing?

Faster execution.

PRACTICAL 4 — BERKELEY ALGORITHM

Aim

Implement Berkeley clock synchronization algorithm.

Simple Theory

Used to synchronize clocks in distributed systems.

Working

1. Master asks time from all systems
2. Calculates average
3. Sends adjustment values
4. All clocks synchronize

Example

System	Time
P1	10:00
P2	10:02
P3	09:58

Average:

10:00

Viva Questions

Q1. Why synchronization needed?

To maintain consistent timing.

Q2. What is Berkeley algorithm?

Centralized clock synchronization algorithm.

Q3. Does Berkeley use UTC?

No.

Q4. Who calculates average?

Master process.

Q5. Advantage of Berkeley algorithm?

Simple and efficient.

PRACTICAL 5 — TOKEN RING MUTUAL EXCLUSION

Aim

Implement token ring mutual exclusion algorithm.

Simple Theory

Processes arranged in ring.

Only process having token can enter critical section.

Working

P1 → P2 → P3 → P4

Token circulates continuously.

Critical Section

Shared resource area.

Code Logic

Java

Meaning:

Only token holder enters critical section.

Viva Questions

Q1. What is mutual exclusion?

Only one process accesses shared resource at a time.

Q2. What is token?

Permission to enter critical section.

Q3. Advantage of token ring?

No starvation.

Q4. Disadvantage?

Token loss problem.

Q5. What is deadlock?

Processes waiting forever.

PRACTICAL 6 — BULLY & RING ELECTION ALGORITHM

A) Bully Algorithm

Theory

Highest process ID becomes coordinator.

Working

If coordinator fails:

- lower process sends election message
- highest ID wins

Example

P1 P2 P3 P4

P4 becomes coordinator.

Viva Questions

Q1. Why called Bully algorithm?

Highest ID process dominates others.

Q2. What is election algorithm?

Algorithm to select coordinator.

Q3. Advantage of Bully?

Fast election.

Q4. Disadvantage?

More messages.

B) Ring Algorithm

Theory

Processes arranged in logical ring.

Election message circulates.

Highest ID wins.

Viva Questions

Q1. Difference between Bully and Ring?

Bully

Ring

More messages

Less messages

Faster

Simpler

Q2. What happens if coordinator crashes?

Election starts again.

PRACTICAL 7 — WEB SERVICE

Aim

Create web service and consume it.

Simple Theory

Web service allows communication between applications over network.

Types

Type	Format
SOAP	XML
REST	JSON/XML

REST Methods

Method	Purpose
GET	Read
POST	Insert
PUT	Update
DELETE	Delete

Simple Flow

Client → HTTP Request → Server → Response

Viva Questions

Q1. What is web service?

Software service accessible over network.

Q2. What is REST API?

Web service using HTTP methods.

Q3. Difference between SOAP and REST?

SOAP

REST

Heavy

Lightweight

XML only

JSON/XML

Q4. What is WSDL?

Web Service Description Language.

Q5. Advantage of REST?

Fast and lightweight.

PRACTICAL 8 — MINI PROJECT

Multiplayer Game Distributed Application

Important Viva Questions

Q1. What is your project?

Explain in:

- problem statement
- technology
- modules
- output

Q2. Why distributed system used?

Supports multiple users simultaneously.

Q3. What protocol used?

Usually TCP/IP or WebSocket.

Q4. What challenges faced?

- synchronization
- network delay
- concurrency

Q5. Future scope?

- cloud deployment
- real-time chat
- security improvements

COMMON EXTERNAL VIVA QUESTIONS

Q1. What is distributed system?

Collection of independent systems working together.

Q2. Advantages?

- Scalability

- Reliability
- Resource sharing
- Fault tolerance

Q3. What is scalability?

Ability to handle increasing workload.

Q4. What is transparency?

Hides distributed nature from users.

Q5. Types of transparency?

- Access
- Location
- Replication
- Failure

Q6. What is fault tolerance?

System continues working after failure.

Q7. Difference between centralized and distributed systems?

Centralized

Distributed

Single server

Multiple nodes

Less reliable

More reliable

Q8. What is concurrency?

Multiple processes executing together.

Q9. What is synchronization?

Coordination between processes.

Q10. What is middleware?

Software layer between client and server.

MOST IMPORTANT DEFINITIONS

Thread

Lightweight process.

Process

Program in execution.

Deadlock

Processes waiting forever.

Starvation

Process never gets resource.

RPC

Remote Procedure Call.

Distributed Computing

Multiple computers solving problem together.

MOST IMPORTANT PRACTICAL COMMANDS

RMI

```
javac *.java
rmiregistry
java RMIServer
```

Bash

CORBA

```
idlj -fall Calculator.idl
javac *.java
java Server
```

Bash

MPI

```
mpicc sum.c
```

Bash

VERY SHORT 1-MARK ANSWERS

Question	Answer
Full form of RMI	Remote Method Invocation
Full form of CORBA	Common Object Request Broker Architecture
Full form of MPI	Message Passing Interface
Full form of ORB	Object Request Broker
Token used for	Mutual exclusion
Bully algorithm selects	Highest ID process



Question

REST uses

SOAP uses

Answer

HTTP

XML

 **Sources**